Michael Ramirez

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Machine Learning Engineer on the OCR and state classification task heads of a multi-task license plate recognition model deployed nationwide, with additional work in edge deployment and inference optimization for real-time video analytics. Focus areas: failure-mode analysis, data drift detection, and targeted dataset curation and fine-tuning to hold model accuracy against production thresholds. Independent research background in transformers, reinforcement learning, and reproducible experimental design.

WORK EXPERIENCE

Flock Safety Atlanta, GA
Machine Learning Engineer, Multimodal ML *October 2025 – Present*

- Develop and evaluate the license plate OCR and license plate state classification heads of the production license plate recognition model, covering head-level training, evaluation protocol, and error analysis.
- Diagnose customer-reported misreads by isolating character-confusion patterns and unfamiliar plate fonts, tracing them to distribution shift in production data the deployed model was never trained against.
- Drove edge-case plate categories from roughly 100% error on 100-plate sampled audits down to under 10% through targeted dataset curation and fine-tuning, measured by per-category F1 on held-out samples.
- Monitor OCR and state prediction accuracy nationwide across all U.S. states and day/night capture conditions, investigating any category that drops below threshold and driving retraining to restore it.
- Benchmark the production model against third-party OCR and image-quality models from Hugging Face, running multiple architectures in parallel to surface model weaknesses and validate training data quality.
- Run and monitor training pipelines through the platform team's Prefect orchestration workflows, tracking experiment progress and diagnosing failed runs, and partner with the ML tooling team on workflow improvements.

Marani Health Oakdale, MN
Jr. Data Scientist *May 2024 – June 2025*
Data Science Intern *May 2023 – May 2024*

- Sole data scientist at the company, leading all data science and AI development for a HIPAA-compliant maternal health platform spanning data infrastructure, clinical trial analysis, model development, and LLM fine-tuning.
- Built and maintained the data lake and ingestion pipelines consolidating fragmented clinical and device data into a single analysis-ready source, and produced regulatory-grade statistical reporting for clinical trial research.
- Built and fine-tuned patient-facing LLM features, including clinical copilots integrated with SQL pipelines and prompt engineering workflows, cutting provider review time for patient summaries and survey insights.
- Deployed production AI services on AWS and SageMaker using FastAPI and asynchronous job queues, maintaining secure logging and HIPAA compliance.

RESEARCH & PROJECTS

Multi-Agent Spatio-Temporal Transformer for Receiver Openness | Georgia Tech 2026

- Built a 1.3M-parameter transformer with spatial and temporal attention and self-supervised trajectory pretraining; reached 77.4% top-1 against a 79.6% zero-parameter constant-velocity baseline, statistical parity on held-out weeks (paired McNemar $p = 0.54$).
- Ran validity work that retracted the original 82–84% course result, tracing it to a leaked label, a padding bug feeding the baseline zeros, and a pipeline that never loaded its data; all three are now CI regression tests verified to fail when the bug is reintroduced.
- Every reported number regenerates from committed artifacts via one make target, with full code and data public. Manuscript targeting MIT Sloan Sports Analytics Conference 2027. github.com/BattleTaco/nfl-receiver-openness

Non-Invasive Fossil Prospecting from GPR | Independent Research 2026 – Present

- Sole author. Formulated and tested a falsifiable hypothesis that void-trained subsurface detectors fail on fossil-like targets because reflection polarity inverts for high-permittivity media.
- Built a controlled gprMax benchmark varying only target dielectric, plus a zero-parameter matched-filter physics baseline that separates all four conditions including a magnitude-matched anti-bone control; YOLOv8n reaches mAP@0.5 of 0.58–0.60 on synthetic B-scans. github.com/BattleTaco/paleo-gpr-ml

EDUCATION

Georgia Institute of Technology, M.S. Computer Science Jan 2025 – Expected May 2028
University of Minnesota – Twin Cities, B.A. Computer Science May 2024

Graduate coursework: Deep Learning, Reinforcement Learning, Machine Learning, Machine Learning for Trading, AI Ethics

SKILLS

Python, PyTorch, TensorFlow, Scikit-learn, SQL, PostgreSQL, Computer Vision, OCR, Multi-Task Learning, NLP, LLM Fine-Tuning, Reinforcement Learning, Hugging Face, Model Evaluation, Data Drift Detection, Edge Deployment, Inference Optimization, Pandas, NumPy, AWS, SageMaker, Snowflake, FastAPI, Docker, MLOps, Prefect, CI/CD, Linux, Git. Bilingual: English, Spanish